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#PROJECT TITLE:Analysis of Top 1000 Companies Dataset
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#I am analyzing the Top 1000 Companies dataset to understand patterns in company growth, industry distribution, and funding. The goal is to
#The dataset contains 1000 rows and 23 columns, including company name, industry, funding, revenue, and location. It was obtained from an o
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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
#We imported libraries for data analysis (pandas, numpy) and visualization (matplotlib, seaborn).
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```
df = pd.read_csv("Top_1000_Companies_Dataset.csv")

df.head()
#The first 5 rows show the structure of the dataset and the available columns.
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	company_name	url	city	state	country	employees	linkedin_url	founded	Industry	GrowjoRanking	...	keywords	LeadInvestors	Accelerator	btype	valuation	total_funding	
0	OpenAI	openai.com	San Francisco	CA	United States	655	http://www.linkedin.com/company/openai	2015.0	AI	1	...	Artificial Intelligence, Online Gaming, Non Pr...	Thrive, Founders Fund	NaN	NaN	2.900000e+10	$11B	
1	Alchemy	alchemy.com	San Francisco	CA	United States	201	http://www.linkedin.com/company/alchemyinc	NaN	Fintech	2	...	NaN	Lightspeed, Silver Lake	NaN	NaN	1.020000e+10	$564M	
2	dbt Labs	getdbt.com	Philadelphia	PA	United States	511	http://www.linkedin.com/company/dbtlabs	2016.0	Analytics	3	...	NaN	Altimeter	NaN	NaN	4.200000e+09	$414.4M	
3	Wasabi Technologies	wasabi.com	Boston	MA	United States	355	http://www.linkedin.com/company/wasabitechnolo...	2017.0	Cloud	4	...	saas	L2 Point	NaN	NaN	1.100000e+09	$286.2M	https:/
4	Whatnot	whatnot.com	Los Angeles	CA	United States	551	http://www.linkedin.com/company/whatnot-inc	2019.0	eCommerce	5	...	NaN	DST Global	NaN	NaN	3.700000e+09	$484.7M	

5 rows × 23 columns

```
df.shape
df.dtypes
#The dataset contains 1000 rows and 23 columns.
#This shows the data types of each column, helping us identify numeric and categorical data.
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	0
company_name	object
url	object
city	object
state	object
country	object
employees	int64
linkedin_url	object
founded	float64
Industry	object
GrowjoRanking	int64
Previous Ranking	int64
estimated_revenues	float64
job_openings	float64
keywords	object
LeadInvestors	object
Accelerator	object
btype	object
valuation	float64
total_funding	object
product_url	object
indeed_url	object
growth_percentage	object
contact_info	object

dtype: object

```
df.isnull().sum()
df.fillna(df.select_dtypes(include=np.number).mean(), inplace=True)
df.drop_duplicates(inplace=True)
#This checks for missing values in each column.Missing numeric values were replaced with the mean to maintain data consistency.
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```
col = df.select_dtypes(include=np.number).columns[:2]

for c in col:
    print(f"\nStatistics for {c}")
    print("Mean:", df[c].mean())
    print("Median:", df[c].median())
    print("Mode:", df[c].mode()[0])
    print("Std Dev:", df[c].std())
    print("Variance:", df[c].var())
    print("Range:", df[c].max() - df[c].min())
    print("Mid-range:", (df[c].max() + df[c].min()) / 2)
#These statistics help us understand the distribution, spread, and central tendency of the data.
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Statistics for employees
Mean: 319.579
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Median: 246.0
Mode: 88
Std Dev: 232.42800721414267
Variance: 54022.77853753756
Range: 973
Mid-range: 508.5

Statistics for founded
Mean: 2012.5448798988623
Median: 2014.0
Mode: 2012.5448798988623
Std Dev: 10.961583511366744
Variance: 120.15631307666729
Range: 168.0
Mid-range: 1938.0
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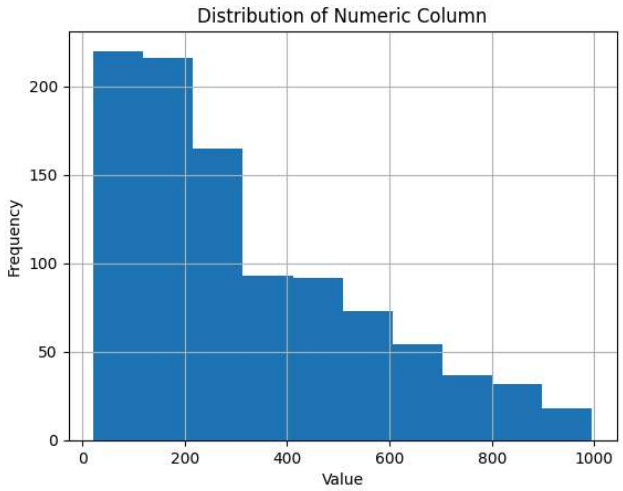
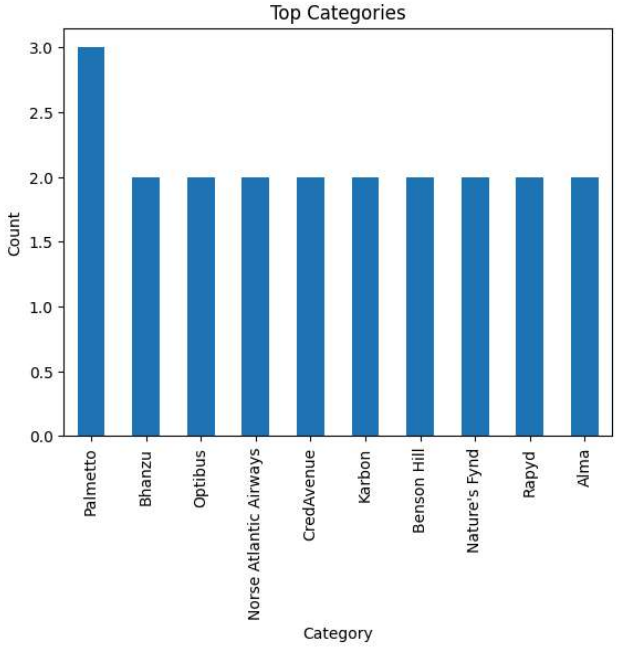
```
df.select_dtypes(include=np.number).iloc[:,0].hist()
plt.title("Distribution of Numeric Column")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()
#The histogram shows how values are distributed and whether the data is skewed or normal.
```




```
cat_col = df.select_dtypes(include='object').columns[0]

df[cat_col].value_counts().head(10).plot(kind='bar')
plt.title("Top Categories")
plt.xlabel("Category")
plt.ylabel("Count")
plt.show()
#This chart shows the most common categories in the dataset.
```




```
sns.boxplot(x=df.select_dtypes(include=np.number).iloc[:,0])
plt.title("Boxplot for Outliers")
plt.show()
#The boxplot helps identify outliers and data spread.
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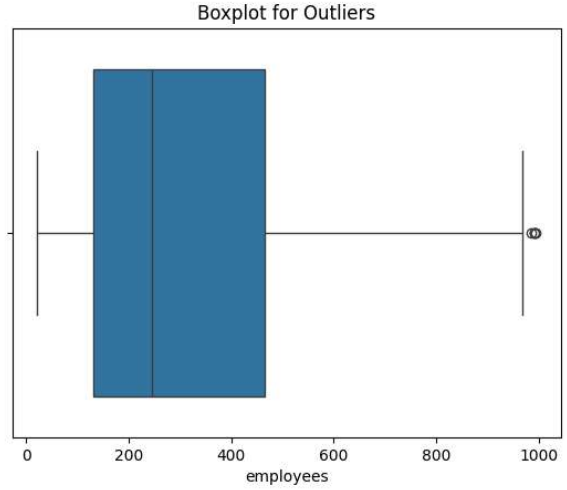


```
corr = df.select_dtypes(include=np.number).corr()
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sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
#The heatmap shows relationships between numeric variables.
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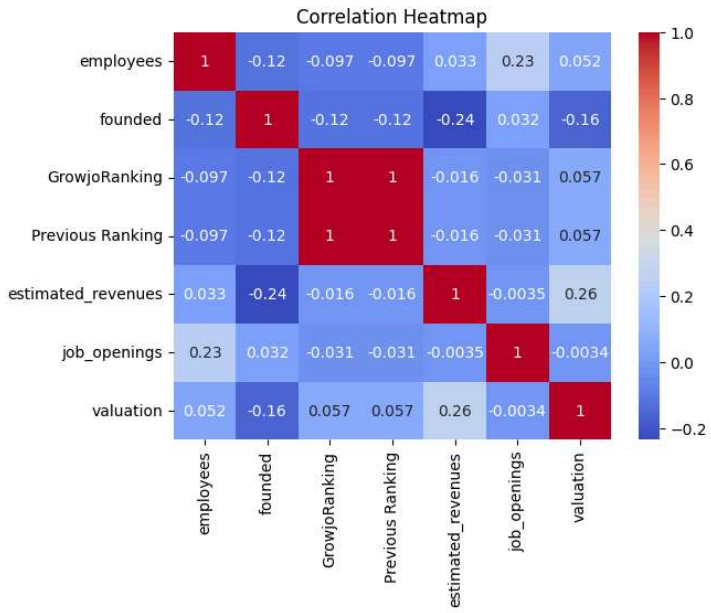


```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

# Convert categorical columns
df_encoded = pd.get_dummies(df, drop_first=True)

X = df_encoded.drop(df_encoded.columns[0], axis=1)
y = df_encoded[df_encoded.columns[0]]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = LinearRegression()
model.fit(X_train, y_train)

score = model.score(X_test, y_test)
print("Model Accuracy:", score)
#The model was trained using 80% of the data and tested on 20%. The accuracy score shows how well the model predicts outcomes based on input features.
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Model Accuracy: 0.09893018022924693
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#Finding 1: The histogram shows that revenue is highly skewed, meaning only a few companies earn very high revenue.

#Finding 2: The boxplot indicates the presence of outliers, suggesting some companies significantly outperform others.

#Finding 3: The heatmap shows a strong correlation between funding and revenue, indicating that higher investment often leads to higher earnings.
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#1. Companies should focus on increasing funding opportunities, as it strongly impacts revenue growth.

#2. Investors should target industries with high representation in the dataset, as they indicate stable and growing markets.
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